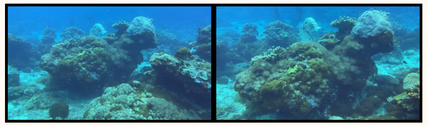
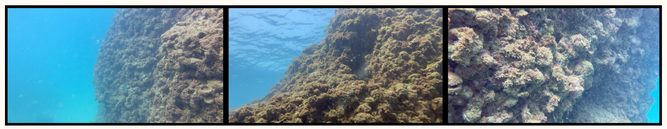
Motion Understanding and Reasoning

Action Hallucination



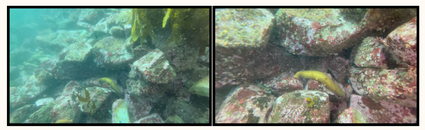
Q: Summarise video?
A: The camera transitions from a medium-wide shot of this marine landscape into a closer view focusing on part of the central object... This smooth, fluid tracking and panning motion gently glides forward and slightly arcs around the environment while keeping the main coral reef structure centered in the frame.
Q: You will evaluate the descriptions on dimension:
Temporal Description: Analyze whether the depiction of the object's motion aligns with the actual movements described in the correct answer. Please assign a score for the dimension, using an integer from 0 to 5
Correct answer:
The camera pans from a wide-angle view of the coral reef to the seahorse-shaped pillars, from close-up to panoramic perspectives.
A: Score: 4. Explanation: The description of the camera movement closely aligns with the ground truth. Both accounts accurately depict a transition from a wider view of the marine landscape to a closer perspective.

Perspective-Dependent Observations



Which camera position or camera movement best matches the video?
(A) The video takes a close-up of the cross; the video is clear; camera angle/movement: horizontal shot, moving from the head down to the tail of the cross
(B) The camera holds a steady wide shot; camera angle/movement: the frame remains fixed
(C) The video takes a wide shot of the school of fish swimming gently around the coral reef; camera angle/movement: the camera shoots vertically to capture the whole coral reef
(D) The camera takes a close-up of hard coral from one spot to another; camera angle/movement: close-up shot around the hard coral

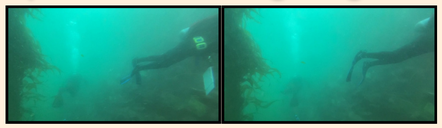
Open-ended with motion understanding



Q: How does the camera's wide horizontal pan (pan left and right across the scene) affect the captured state of the seaweed in the coral reef ecosystem?
A: The camera's wide horizontal sweep captures the seaweed's flexible swaying movement around the rocky reef with clear image quality

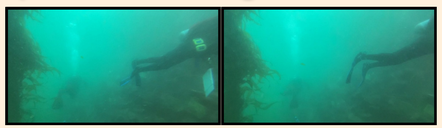
Spatiotemporal Analysis

Spatiotemporal Monitoring - Light



What is the lighting condition in the video?
(A) dim or low-light conditions
(B) medium depth, light reaches the area
(C) sunlight can penetrate through
(D) aphotic zone

Spatiotemporal Monitoring - Environment



How is the environment in the video?
(A) dark blue seawater
(B) dark rocky seabed
(C) green, turbid water
(D) clear water environment, deep blue water

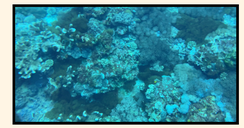
Organism Behavioral Kinematics



Which movement or state of the shark best matches the video?
(A) the shark swims at normal speed
(B) the shark swims fairly fast
(C) the shark swims slowly
(D) the shark swims slowly and exhales air bubbles

Object-centric Understanding

Marine Biotic Communities



What is the central object in the video?
(A) coral reef
(B) sandy seafloor
(C) sea anemone
(D) sea sponge

Phenotypic/ Morphological Traits



Which color and size or shape description best matches the central object?
(A) color: rough body, black and pink; size/shape: various sizes
(B) color: grayish white; size/shape: adult sizes
(C) color: mainly yellow and black, with blue stripes; size/shape: various sizes
(D) color: green, light green, and dark green

Environmental Context Fabrications
Biological Entity Hallucination



Q: Summarise video?
A: An exceptionally clear, sharp, and high-definition underwater scene characterized by a slow, gentle, and deliberate pace. In the initial frames, a massive central coral reef cluster heavily carpeted in dense green and brown algae dominates the view, surrounded by smaller branching formations. This maneuver dynamically reveals a small, patterned pufferfish swimming calmly on the left—whose leisurely movement... as well as a prominent, light-colored artificial seahorse statue fixed to the seabed...

Q: You will evaluate the descriptions on dimension:
1. Subject Correspondence: Assess whether the subject of the input description accurately matches the subject indicated in the ground truth description.
2. Hallucination Detection: Identify any discrepancies by determining if the description includes facts, actions, or elements that do not exist in reality.
Please assign a score for the dimension, using an integer from 0 to 5
Correct answer:
At the bottom of the sea, coral reefs display shades of moss green, light blue, light purple, and yellow. Large, white, seahorse-shaped iron pillars stand tall. The coral reefs and pillars vary in size. The seawater is crystal clear, and the high-quality video captures images of marine life, corals, and the stationary iron pillars.
A:
1. Subject Correspondence: 5. Explanation: The predicted answer perfectly captures the primary subjects outlined in the ground truth.
2. Hallucination Detection: 3. Explanation: The predicted answer introduces details not verified by the ground truth, most notably the presence of a specific "patterned pufferfish swimming calmly on the left" and its explicit movements, whereas the ground truth only makes a general mention of "marine life."